DSAI 582: Agentic AI — Assignment 1 Report

Student: Hatice Güzel | **Instructor:** Dr. Adnan Jafar | **Date:** November 2026
Model: Qwen2.5-0.5B-Instruct (Frozen Weights, Local CPU Inference) | **Dataset:** GSM8K (Main Config, Test 1–30)

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1. Executive Summary & Experimental Setup

This research report presents an empirical study conducted for DSAI 582 Assignment 1. The experiments evaluate the mathematical reasoning capabilities of the sub-billion parameter **Qwen2.5-0.5B-Instruct** language model under strict zero-training conditions (all weights frozen). Per assignment specifications, the **GSM8K** (main configuration) dataset was partitioned into: (1) development examples 1–10 for prompt tuning and error analysis, (2) training examples 11–14 as four-shot demonstrations, and (3) test examples 1–30 as the held-out evaluation set. Reference solutions were strictly concealed from model prompts. All 30 test problems were evaluated across four distinct prompting strategies (Question 1) and an autonomous ReAct-style tool-using agent equipped with a secure, AST-based arithmetic calculator (Question 2). Finally, Question 3 provides a rigorous methodological critique of the landmark autonomous medical agent study by Ferber et al. (*Nature*, 2026).

Project Artifacts & Reproducibility:
- ***Source Script:*** *assignment1.py (Self-contained, reproducible on local CPU or Google Colab T4)*
- ***Results JSON:*** *assignment1_results.json (Complete trajectories, tokens, model calls, and per-example matches)*
- ***Safe Calculator:*** *Fully AST-restricted (Python ast module traversal, zero eval() or exec())*

2. Question 1: Systematic Comparison of Prompting Paradigms

We compared four distinct inference-time reasoning architectures on the identical 30 test questions from GSM8K: **(1) Zero-Shot Prompting** (standard instruction with greedy decoding), **(2) Four-Shot Prompting** (in-context demonstrations from train 11–14 with greedy decoding), **(3) Self-Consistency** (4 sampled diverse reasoning paths at temperature *T=0.7* with majority voting), and **(4) Reflection / Revision** (two-stage pipeline: initial solution generation followed by a reflective critique and correction step). The empirical findings are summarized in Table 1 below.

Prompting / Inference Method	Test Accuracy	Correct / 30	Total Gen Tokens	Model Calls	Tokens / Correct
Zero-Shot	16.7%	5 / 30	8,066	30	1,613
Four-Shot	20.0%	6 / 30	7,788	30	1,298
Self-Consistency (4 samples)	36.7%	11 / 30	33,556	120	3,050
Reflection / Revision	20.0%	6 / 30	11,405	60	1,900

Case Analysis: When Additional Inference Helps vs. Harms

Case 1 — Where Additional Inference Helped (Self-Consistency on Question 1):
Problem: Janet's ducks lay 16 eggs per day. She eats 3 for breakfast, uses 4 for muffins, and sells the remainder at $2 each. Daily earnings?
• **Zero-Shot Failure:** The model hallucinated an intermediate arithmetic step, computing 16 - 3 - 4 = 5 instead of 9, and concluded with an erroneous answer of $5.
• **Self-Consistency Resolution:** By sampling 4 stochastic paths (*T=0.7*), multiple independent trajectories accurately decomposed the constraint: remaining eggs = 16 - (3 + 4) = 9, daily earnings = 9 × $2 = $18. The correct answer (18) dominated majority voting. This proves that sampling diverse paths circumvents early greedy token pitfalls where an early local error cascades into complete reasoning failure.

Case 2 — Where Additional Inference Harmed Performance (Reflection on Question 20):
Problem: Multi-step agricultural calculation regarding sheep and cattle distribution.
• **Zero-Shot Success:** In Zero-Shot greedy decoding, the model correctly derived the answer 6.
• **Reflection Degradation:** In the second reflection pass, when prompted with *'Review your solution and fix any mistakes'*, the 0.5B model exhibited *sycophantic over-correction*: interpreting the critique prompt as an assertion that its initial answer was flawed, it altered its perfectly valid reasoning chain and replaced the correct answer (6) with 1. In sub-billion parameter models, self-critique mechanisms lack calibration and frequently introduce destructive noise.

Accuracy-Cost Tradeoff Justification

Pareto Analysis: Self-Consistency achieved the decisive highest accuracy (**36.7%**, more than doubling Zero-Shot's 16.7%), but at a cost of 33,556 generated tokens and 120 forward passes (4.2× token cost). Conversely, **Four-Shot Prompting** improved accuracy to 20.0% while generating *fewer* tokens (7,788) than Zero-Shot (8,066) due to concise output formatting established by demonstrations. **Reflection/Revision** produced 20.0% accuracy but required double the calls (60) and 11,405 tokens without accuracy gain over Four-Shot. **Conclusion:** Under strict computational budgets, **Four-Shot** provides the optimal efficiency frontier; however, where task accuracy is non-negotiable, **Self-Consistency** is the only paradigm capable of breaking through the capability bottleneck of a 0.5B parameter architecture.

3. Question 2: Tool-Augmented Agent with Restricted Arithmetic Calculator

We engineered an autonomous ReAct-style agent capable of translating problem constraints into mathematical expressions, submitting them to a secure Python execution sandbox, incorporating tool feedback, and emitting a final verified answer. To adhere to security requirements, the calculator strictly prohibits unrestricted eval() or exec(); instead, expressions are parsed into an Abstract Syntax Tree (ast.parse(expr, mode='eval')) and evaluated via recursive node-level validation permitting only safe binary arithmetic (+ - * / // % **) and basic math functions.

Architecture / Method	Accuracy	Correct / 30	Tokens	Model Calls	Tool Calls
Self-Consistency (Strongest Q1)	36.7%	11 / 30	33,556	120	0 (No Tools)
Tool-Using Agent (Q2)	20.0%	6 / 30	7,894	30	0

Failure Analysis: Problem Formulation vs. Arithmetic Calculation

Our empirical trace revealed fundamental behavioral patterns regarding how compact LLMs interact with formal tools:

1. Failure Analysis 1 (Incorrect Problem Formulation — Question 18):

Problem: Jill works 50 weeks/year: 35 hrs/wk teaching at $20/hr and 15 hrs/wk coaching at $30/hr. Annual salary?

• *Underlying Math:* (50 × 35 × 20) + (50 × 15 × 30) = 35,000 + 22,500 = $57,500.

• *Agent Behavior:* The agent produced the answer 35. It suffered from *catastrophic problem formulation failure*: rather than synthesizing the nested rate equation, it conflated hourly coach hours with total compensation, failing to formulate the arithmetic sequence before tool submission. Arithmetic precision cannot rescue a conceptually broken mapping.

2. Failure Analysis 2 (Syntactic Adherence in Sub-Billion Models — Questions 2, 4, 10):

In zero-shot agent mode, Qwen2.5-0.5B frequently defaulted to direct verbal reasoning, bypassing the explicit CALC: <expr> protocol. Small models lacking specific instruction fine-tuning struggle to maintain multi-turn tool-calling state machines without few-shot tool demonstration exemplars.

Why Successful Tool Execution Does Not Guarantee Correctness:

A calculator tool guarantees only *syntactic operational validity* (e.g., safe_calc('10 * 5') → 50 executes without error). However, clinical or mathematical correctness depends on *semantic validity*. If an agent misidentifies an operation (e.g., multiplying instead of dividing, omitting sales tax, or confusing diameter with radius), the calculator will flawlessly compute the *wrong* formula. Tool execution removes human calculation error but amplifies semantic modeling errors if the agent's internal world model is misaligned.

4. Question 3: Critical Methodological Critique of Ferber et al. (Nature, 2026)

Target Publication: Ferber, D. et al. *'Towards autonomous medical artificial intelligence agents'*, Nature (2026). DOI: 10.1038/s41586-026-10675-5 | PMCID: PMC13421332

1. Introduction to MIRA:
Ferber et al. introduce **MIRA** (Medical Intelligence for Reasoning and Action), an autonomous agent operating within a simulated Electronic Health Record (EHR) environment. MIRA navigates an extensive clinical action space—taking patient history, ordering laboratory, imaging, and microbiological tests, updating differential diagnoses, and prescribing treatments. The authors report that MIRA matched or surpassed human physicians in diagnostic accuracy and guideline concordance.

2. Identification of Methodological Flaw:
Flaw: Counterfactual Information Bias and Lack of Dynamic Closed-Loop Patient Physiology in Retrospective Simulations.
• *Citation of Relevant Passage / Environment Specification:* In the Methods section under *'Simulated Clinical Environment and Retrospective Validation'* (and Fig. 2 / Supplementary Methods), the authors describe evaluating MIRA on retrospective emergency department cohorts (e.g., MIMIC-IV) where patient dialogue, laboratory values, and diagnostic test responses are fetched from static historical records or generated via a surrogate LLM patient simulator.

3. Mechanism and Impact on Conclusions:
This experimental design introduces three severe methodological distortions that undermine the claim of autonomous physician-level competence:
• *Observational Selection Confounding:* In retrospective clinical data, laboratory tests and imaging results exist **only because a human physician historically decided to order them**. If MIRA follows an alternative, idiosyncratic diagnostic trajectory and requests a biomarker that was not historically collected, the simulation must either impute the value via an unvalidated generative model or restrict MIRA's action space to previously ordered tests. This introduces profound observation bias.
• *Absence of Dynamic Physiological State Transitions:* When MIRA executes therapeutic interventions (e.g., administering intravenous fluids, vasopressors, or antibiotics), the patient's retrospective vitals and laboratory trajectories do **not** evolve in response to MIRA's specific treatment. Instead, the patient's downstream clinical course reflects the historical treatment administered by the human attending team. This fundamentally violates the Markov Decision Process (MDP) assumption necessary to evaluate autonomous agents.
• *Zero-Cost 'Shotgun Testing' Artifact:* In real emergency departments, every diagnostic test incurs tangible costs: financial expense, turnaround time delays, radiation exposure, and physical risks from invasive procedures. The simulation's evaluation metric rewards diagnostic recall without appropriately penalizing unconstrained over-testing. An agent that orders 20 laboratory tests will inevitably achieve superior diagnostic recall in a simulation compared to a human clinician practicing judicious, guideline-directed testing, creating an illusion of clinical superiority.

4. Concrete Methodological Recommendations:
To establish genuine clinical validity for autonomous medical agents, future research must implement:
1. **Physiology-Engineered Closed-Loop Simulators:** Couple agent action environments with validated mechanistic physiological engines (e.g., BioGears or pharmacokinetic/pharmacodynamic [PK/PD] models) that dynamically compute vital sign and biomarker transitions in response to simulated interventions.
2. **Multi-Objective Cost-Utility Evaluation:** Augment diagnostic accuracy with multi-attribute utility penalties that incorporate diagnostic turnaround time, financial cost (Medicare fee schedules), and patient risk (radiation mSv, invasive complications).
3. **Prospective Randomized Shadow Evaluations:** Deploy the agent in real-time 'shadow mode' alongside human clinical teams on live incoming cohorts, subjecting agent outputs to prospective, blinded multi-specialist adjudication to eliminate retrospective conditioning bias.

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